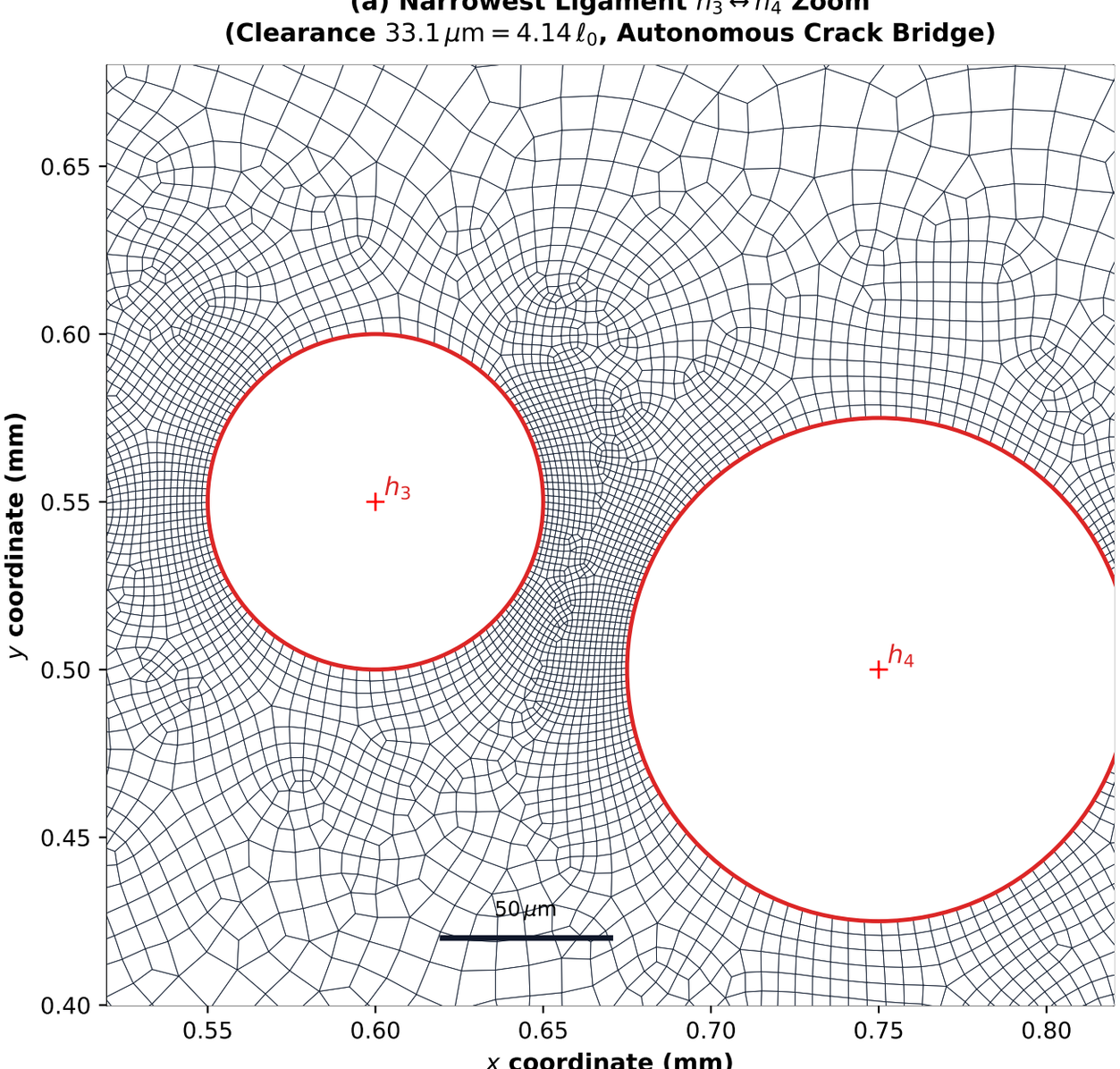
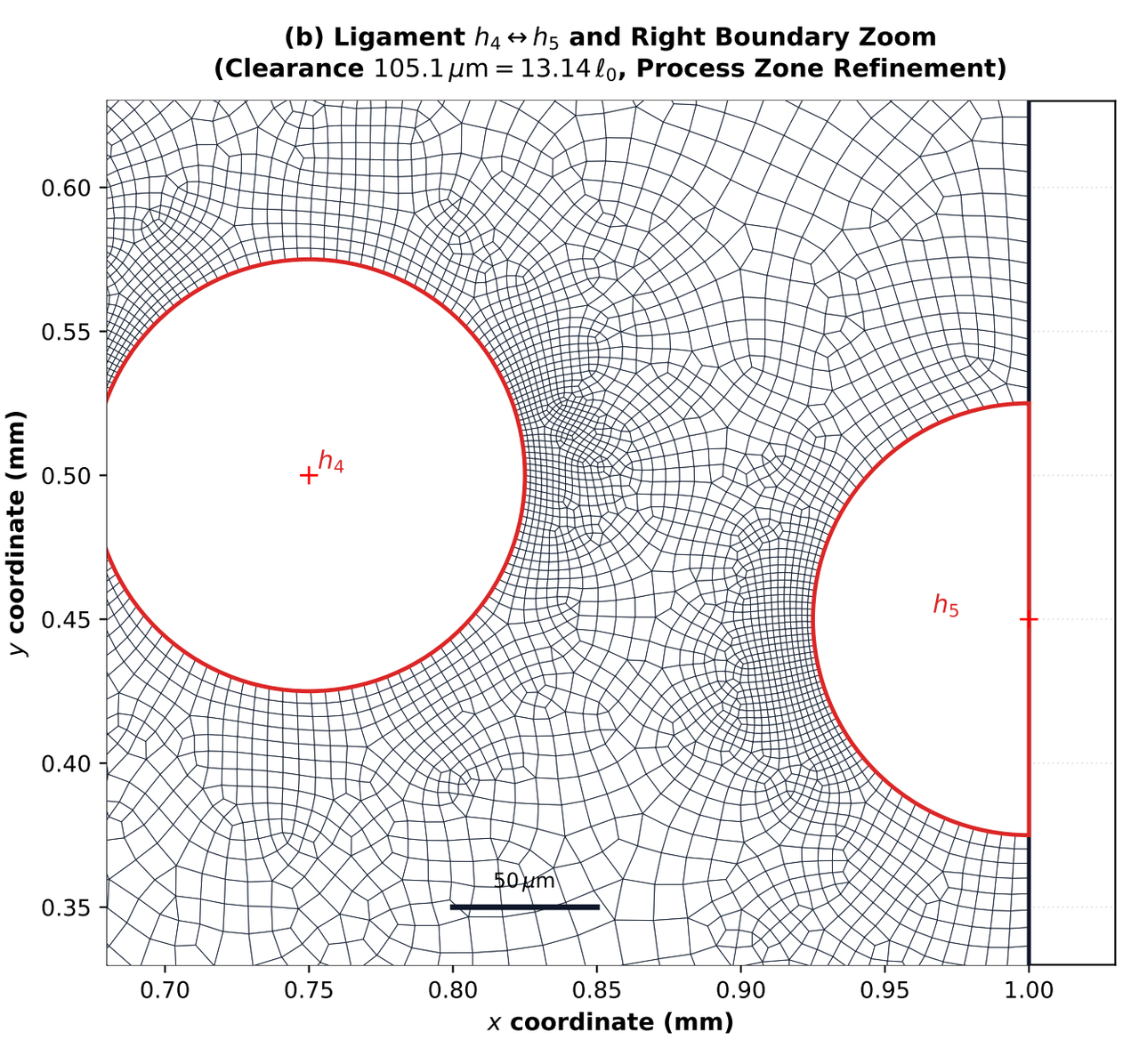

(a) Narrowest Ligament $h_3 \leftrightarrow h_4$ Zoom
(Clearance 33.1 μm = 4.14 ℓ_0, Autonomous Crack Bridge)
h_3
h_4
50 μm
y coordinate (mm)
x coordinate (mm)
0.65
0.60
0.55
0.50
0.45
0.40
0.55
0.60
0.65
0.70
0.75
0.80


(b) Ligament $h_4 \leftrightarrow h_5$ and Right Boundary Zoom
(Clearance 105.1 μm = 13.14 ℓ_0, Process Zone Refinement)
h_4
h_5
50 μm
y coordinate (mm)
x coordinate (mm)
0.60
0.55
0.50
0.45
0.40
0.35
0.70
0.75
0.80
0.85
0.90
0.95
1.00